

E-Commerce Cohort, Funnel & A/B Testing Pipeline

Moving from Descriptive Insights to Experimental, Decision-Ready Analytics

Core Tech: Python (pandas, scipy, Faker) • MySQL (Window Functions, Views) • Power BI Dashboard

1. Executive Summary & Problem Statement

In modern business analytics, **descriptive dashboards** (showing total revenue, sales by category, or basic demographic charts) only explain *what* is happening in a business. They serve as **lagging indicators**—revealing revenue loss or customer attrition after it has already occurred.

However, descriptive analytics fails to answer two critical operational questions:

- 1. **Why and where are customers dropping off** along their journey? (Diagnostic Analytics)
- 2. **Will a proposed product change actually fix the issue** or improve conversion before full deployment? (Experimental Analytics)

Project Mission

This project builds an end-to-end analytics and experimentation pipeline that diagnoses friction points across time (Cohort Analysis) and user journey stages (Funnel Analysis), and statistically validates a targeted product intervention using a randomized A/B test (χ^2 hypothesis testing).

2. Business Value & Motivation ("Why We Are Doing This")

Moving beyond passive reporting builds high-value decision-support systems for e-commerce organizations. Here is the rationale behind each component:

Analytical Pillar	Business Purpose	Key Outcome
1. Cohort Retention Analysis	Tracks repeat customer behavior across months relative to their first purchase date.	Identifies long-term customer decay rates and evaluates baseline customer lifetime retention.
2. Funnel Conversion Analysis	Measures step-to-step drop-offs from initial site visit to final purchase.	Pinpoints exact friction bottlenecks in the user experience (e.g., signup $\rightarrow$ activation).
3. Controlled A/B Testing	Evaluates a newly proposed UI/UX feature against the baseline experience in a randomized trial.	Replaces intuition with statistical rigor (χ^2 significance) before deploying changes to 100% of users.

Interview Elevator Pitch (30 Seconds)

"In this project, I transitioned an e-commerce platform from passive descriptive reporting to diagnostic and experimental analytics. I built SQL views to map monthly cohort decay and front-end funnel drop-offs, then conducted a Python-driven Chi-Square A/B test to statistically prove whether a proposed checkout optimization actually boosted conversion rates."

3. Dataset Architecture & Synthetic Data Strategy

A core highlight of this project is its **hybrid data architecture** combining real historical transactional data with programmatically generated event stream data.

Base Real Dataset: Olist Brazilian E-Commerce (Kaggle)

Contains real multi-month e-commerce transactions from Brazil (`customers`, `orders`, `order\_items`, `payments`). **Why it's essential:** It provides authentic purchase dates and customer IDs spread across multiple years, which is required for accurate Cohort Retention Analysis.

Why Do We Need to Create 2 Synthetic Data Tables?

In real e-commerce companies, front-end clickstream events (Google Analytics/Mixpanel) are stored separately from back-end database orders. Public datasets almost never include clickstreams or experiment logs due to privacy and business confidentiality.

Synthetic Table	Key Fields Generated	Why Creation Was Mandatory
1. Events / Sessions Table (`events`)	`user_id`, `event_type` (signup, activated, purchased), `timestamp`, `channel`	Olist only tracks completed purchases. To calculate funnel drop-offs (e.g. signup $\rightarrow$ activated), we must synthesize pre-purchase touchpoints.
2. A/B Test Table (`ab_test_assignment`)	`user_id`, `group` (control vs treatment), `converted` (0/1)	Public datasets lack controlled experiment metadata. We programmatically bake in an ~8-10% lift in treatment to simulate a true product effect.

Interview Talking Point: Defending Synthetic Data

"When an interviewer asks why I created synthetic data, I explain: 'Real transactional databases like Olist only log completed receipts. To analyze top-of-funnel drop-offs and evaluate an A/B experiment, clickstream and variant assignment tables are required. Simulating these tables using Python's Faker while linking them to real customer keys mirrors modern production enterprise architectures.'"

4. Step-by-Step Execution Workflow Procedure

1 Data Ingestion & Cleaning (Python / Pandas / MySQL)

Load Olist CSVs into Pandas, fix data types (`datetime` parsing), deduplicate records, remove null customer IDs, and push cleaned tables into a MySQL database using `SQLAlchemy`.

2 Synthetic Funnel & Experiment Data Generation

Write a Python script utilizing `Faker` to generate 50,000+ user session events and assign users randomly to `control` or `treatment`. Bake in a deliberate ~8-10% higher purchase conversion rate for treatment to ensure the hypothesis test discovers a real effect.

3 Cohort Retention Engine in SQL (Window Functions & Views)

Write a SQL query using `DENSE_RANK()` and `PERIOD_DIFF()` to group customers by first purchase month (cohort) and track repeat purchases in Month 1, 2, 3... Save as a reusable MySQL view: `vw_cohort_retention`.

4 Funnel Conversion Engine in SQL (Step-to-Step Drop-offs)

Query the `events` table using conditional aggregations to count users at each stage (`signup` $\rightarrow$ `activated` $\rightarrow$ `purchased`). Compute step conversion % and drop-off rates. Save as view: `vw_funnel_conversion`.

5 Statistical A/B Testing in Python (Chi-Square Test)

Pull assignment and conversion outcomes into Pandas. Execute a **Pearson Chi-Square (χ^2) Test of Independence** via `scipy.stats.chi2_contingency` to compare control vs treatment conversion rates. Output p-value, sample size, lift %, and statistical significance.

6 Power BI Dashboard Integration

Connect Power BI directly to MySQL views. Build Page 1: **Cohort Retention Heatmap Matrix** (Month-over-Month decay). Build Page 2: **Funnel Drop-off & A/B Experiment Summary** with KPI cards, slicers for channel/date, and significance status.

5. Key Interview Questions & Answers Cheat Sheet

Q1: Why did you choose a Chi-Square test for the A/B test analysis?

Answer: "Conversion in this context is a binary categorical outcome (1 = converted, 0 = not converted). Since we are comparing proportions across two categorical groups (Control vs Treatment), a Chi-Square test of independence (or a Z-test for two proportions) is the mathematically correct statistical test to determine if the observed difference is significant or random noise."

Q2: How did you compute the cohort index in MySQL?

Answer: "I used a CTE with `MIN(order_purchase_timestamp) OVER (PARTITION BY customer_unique_id)` to capture each user's cohort month. Then I calculated the month index using `PERIOD_DIFF()` between the subsequent order date and the cohort month date, allowing dynamic pivot aggregating in Power BI."

Q3: Why use MySQL Views instead of importing raw tables into Power BI?

Answer: "Pushing complex window aggregations down to the SQL database engine optimizes report performance, reduces Power BI memory load, and ensures a single source of truth that can be re-used by other downstream applications or BI tools."